

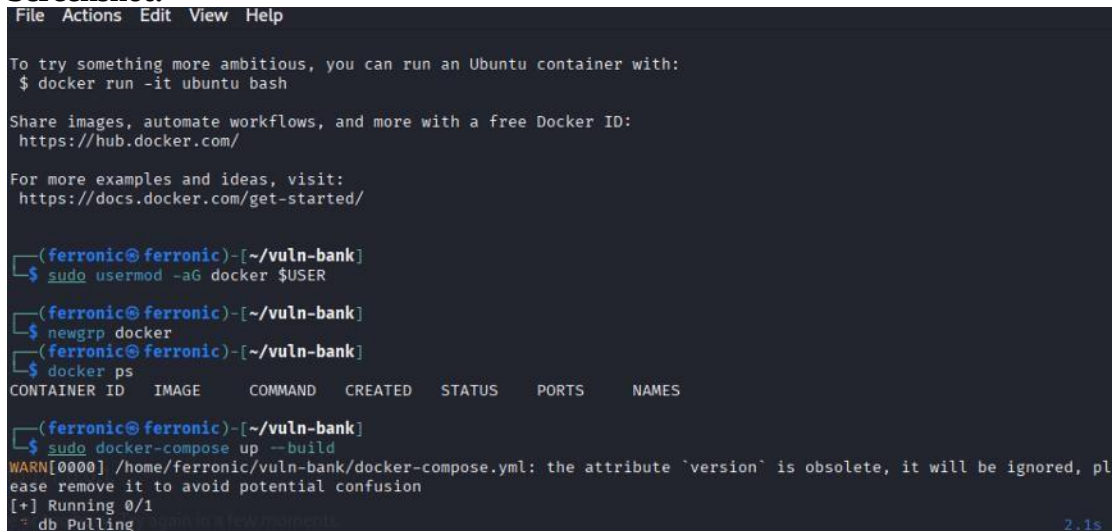
Vulnerability Assessment Report - Vuln-Bank

The Vuln-Bank web application is a deliberately insecure project designed to explore common website security flaws. This report details the evaluation process, where various components such as login, account management, and admin interfaces were analyzed to identify potential weaknesses.

Setup and Exploration

The application was deployed using Docker by cloning the repository from <https://github.com/Commando-X/vuln-bank.git> with the command `git clone https://github.com/Commando-X/vuln-bank.git` && `cd vuln-bank` && `docker-compose up --build -d`. Post-setup, the local site was accessed to initiate testing.

- **Screenshot:**



```
File Actions Edit View Help

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

(ferronic@ferronic)-[~/vuln-bank]
$ sudo usermod -aG docker $USER

(ferronic@ferronic)-[~/vuln-bank]
$ newgrp docker
(ferronic@ferronic)-[~/vuln-bank]
$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
(ferronic@ferronic)-[~/vuln-bank]
$ sudo docker-compose up --build
WARN[0000] /home/ferronic/vuln-bank/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 0/1
  db Pulling    2.1s
```

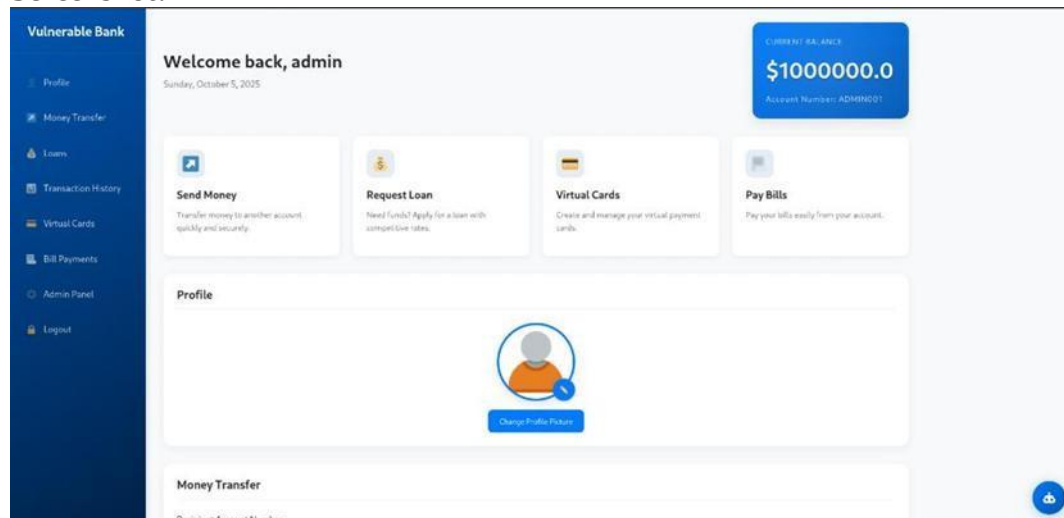
Key Findings

During testing, it was observed that the app fails to validate user inputs adequately, exposing it to malicious actions that could compromise account data.

1. **Login Bypass (SQL Injection)**

- **Issue:** The login accepts inputs like ' OR 1=1 --, bypassing authentication.
- **Impact:** Unauthorized access to admin accounts is possible.

○ **Screenshot:**



2. **Account Modification via Browser**

- **Issue:** Users can alter fields like "is_admin" or "balance" using developer tools.
- **Impact:** Attackers can escalate privileges or inflate balances.

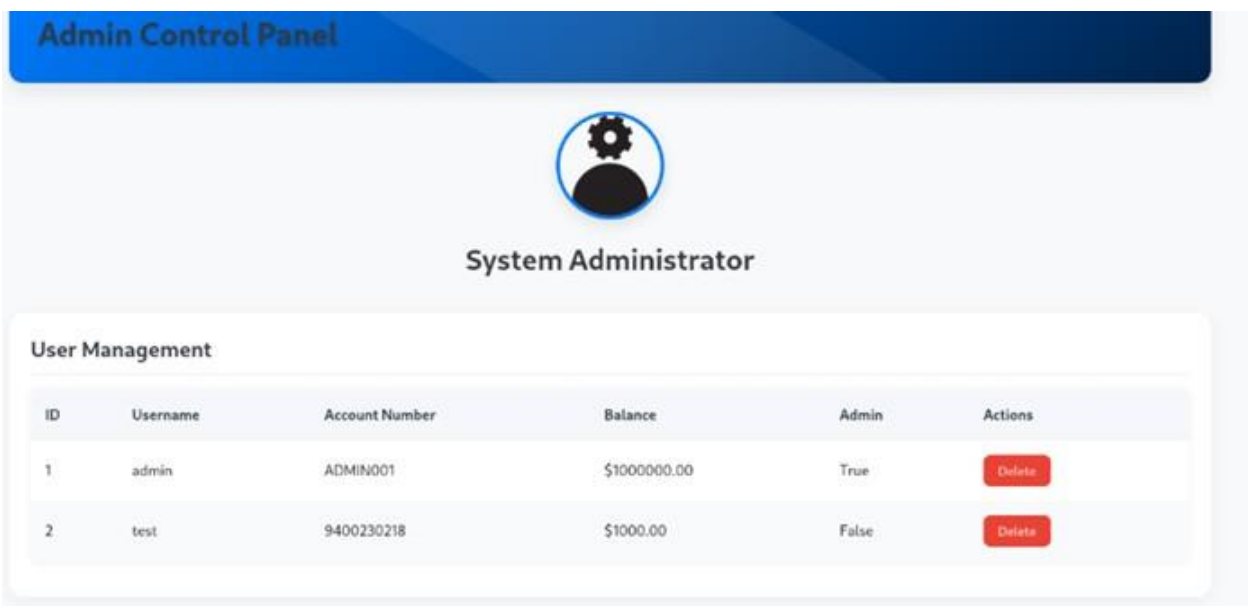
3. **Unsecured Admin Panel**

- **Issue:** Admin pages lack robust access controls.
- **Impact:** Unauthorized users can manipulate sensitive data.

Implications

In a real-world scenario, attackers could exploit these vulnerabilities to:

- Gain admin access without credentials
- Manipulate financial balances
- Manage user accounts
- Access confidential user data



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